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Design and Implementation of an Information System for the Evaluation of Employee Skills and Performance within CNAS

Supervised by:
Mr OMRANI Abdelhak

Prepared by:
BENNADJI Wassim Redha
BENNABDELOUAHAB Ali Ramzi

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ABSTRACT

In a context marked by the digital transformation of public administrations and the modernization of internal processes, human resource management is now a strategic axis essential for improving organizational performance. The National Social Insurance Fund (CNAS), as a large-scale public institution, faces challenges related to the objective evaluation of competencies, the monitoring of individual performance, and the centralization of employee-related data. Within the Human Resources Department (HRD), evaluation procedures still rely on partially manual and fragmented methods, thus limiting the reliability of analyses, information traceability, and the efficiency of management decision-making.

This thesis focuses on the design and implementation of an information system dedicated to the evaluation of employee skills and performance within CNAS. This initiative is part of a digitalization approach aimed at automating the evaluation process, structuring analysis criteria, centralizing HR data, and producing relevant indicators facilitating skills foresight management and strategic decision support.

The methodology adopted is based on a thorough study of the existing situation, a detailed analysis of functional and organizational needs, followed by a modeling phase using UML diagrams, and then a development and implementation phase of a web application compliant with institutional requirements. The proposed system contributes to improving the transparency, efficiency and consistency of the evaluation process, while strengthening human resources governance and the organization's overall performance.

Keywords:

Information System, Human Resources, Performance Evaluation, Skills Management, HRD, Digitalization, CNAS.

ABSTRACT IN ARABIC

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ABSTRACT IN ENGLISH

In the context of digital transformation and the modernization of public administrations, human resource management has become a strategic pillar for improving organizational performance and strengthening institutional governance. The National Social Insurance Fund (CNAS), as a major public institution, faces significant challenges related to the objective evaluation of employee competencies, performance monitoring, and the centralization and reliability of human resources data. This thesis focuses on the design and implementation of an information system dedicated to the evaluation of employee skills and performance within CNAS. The project aims to structure the appraisal process, centralize HR data, improve traceability, and generate meaningful indicators for decision support.

Keywords: Information System, Human Resource Management, Performance Evaluation, Competency Management, HRD, Digital Transformation, CNAS.

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LIST OF ABBREVIATIONS

- CNAS: National Social Insurance Fund.
- HRD: Human Resources Department.
- DMSI: Modernization and Information Systems Department.
- IS: Information System.
- HRMS: Human Resource Management System.
- KPI: Key Performance Indicator.
- DB: Database.
- UML: Unified Modeling Language.
- UI: User Interface.
- RBAC: Role-Based Access Control.
- DPO: Data Protection Officer.
- API: Application Programming Interface.

GENERAL INTRODUCTION

Digital transformation today influences all sectors of economic activity, including public institutions. This evolution is not limited to the simple computerization of administrative tasks; it aims above all at continuous improvement in the quality of services, reduction in processing times, data reliability, and strengthening of decision support mechanisms. In this context, human resource management occupies a strategic position, as it directly impacts work organization, institutional performance, skills development, and overall governance.

The National Social Insurance Fund (CNAS), as a large-scale national public institution, manages significant human capital, distributed across several central departments, regional agencies, and local structures. The quality of benefits provided to members depends largely on internal organizational efficiency and human resources performance. The Human Resources Department (HRD) and the Modernization and Information Systems Department (DMSI) must therefore have reliable, centralized and usable information about employees: competencies, career paths, training completed, evaluations, assigned objectives, results achieved, and development needs.

However, in many organizations, competency and performance evaluation processes still rely on heterogeneous practices: paper documents, scattered files, sometimes non-standardized criteria, and absence of a structured traceability system. These limitations reduce analysis quality, complicate individual monitoring, and slow down the implementation of workforce planning and development management.

It is in this context that our project takes place: **to design and implement an information system dedicated to the evaluation of employee skills and performance within CNAS**. The objective is to propose a digital solution allowing to structure the evaluation process, centralize HR data, and produce key performance indicators useful for strategic management, while ensuring traceability, confidentiality, and security of access.

This thesis is organized as follows:

- **Chapter 1:** Presentation of the general context.
- **Chapter 2:** Analysis and design.
- **Chapter 3:** Implementation.

CHAPTER 1

PRESENTATION OF THE GENERAL CONTEXT

1.1 Introduction

In this chapter, we present the overall framework of our project. We begin by introducing the host organization, CNAS, then describe the Human Resources Department (HRD) and the Modernization and Information Systems Department (DMSI). Next, we analyze the current situation regarding competency and performance evaluation, in order to identify limitations and needs. Finally, we formulate the research problem, objectives, and the proposed solution.

1.2 Presentation of CNAS

1.2.1 Legal Status and General Framework

The National Social Insurance Fund for Salaried Workers (CNAS) is a public administrative body, endowed with legal personality and financial autonomy, pursuant to Law No.° 88-01 of January 12, 1988, relating to public bodies with administrative character. It is also presumed to be commercial in its relations with third parties, allowing it to enter into agreements and manage significant financial flows within the framework of social protection. CNAS is governed by a **Board of Directors** and placed under the supervision of the **Minister of Labor, Employment and Social Security**. Its headquarters is located in Algiers (Ben Aknoun). It has national competence and has central and local services distributed throughout the territory [4].

1.2.2 Main Missions and Attributions

CNAS is one of the main pillars of the social security system in Algeria. Its attributions include the management of social insurance, occupational hazards, family benefits, collection, medical supervision, and conventions with healthcare providers. This diversity of activities shows that the institution depends heavily on the quality of its internal processes and the efficiency of its human capital.

1.2.3 CNAS Organization

CNAS organization is based on a hierarchical and territorial structure that allows it to ensure national coverage. It mainly comprises:

- a **General Directorate**;
- **central services**;
- **59 regional agencies**;
- **768 payment structures**;
- **health and social structures**.

This diversity has a direct consequence for our project: the evaluation system cannot be based on a uniform grid applied identically to all professions. Position families must be distinguished, criteria adapted, and comparisons limited to homogeneous areas to avoid unfair or unusable rankings. This idea aligns with research showing that a relevant evaluation system must be both standardized in its structure and adaptable according to organizational context, hierarchical levels, and job families [2, 10, 1].

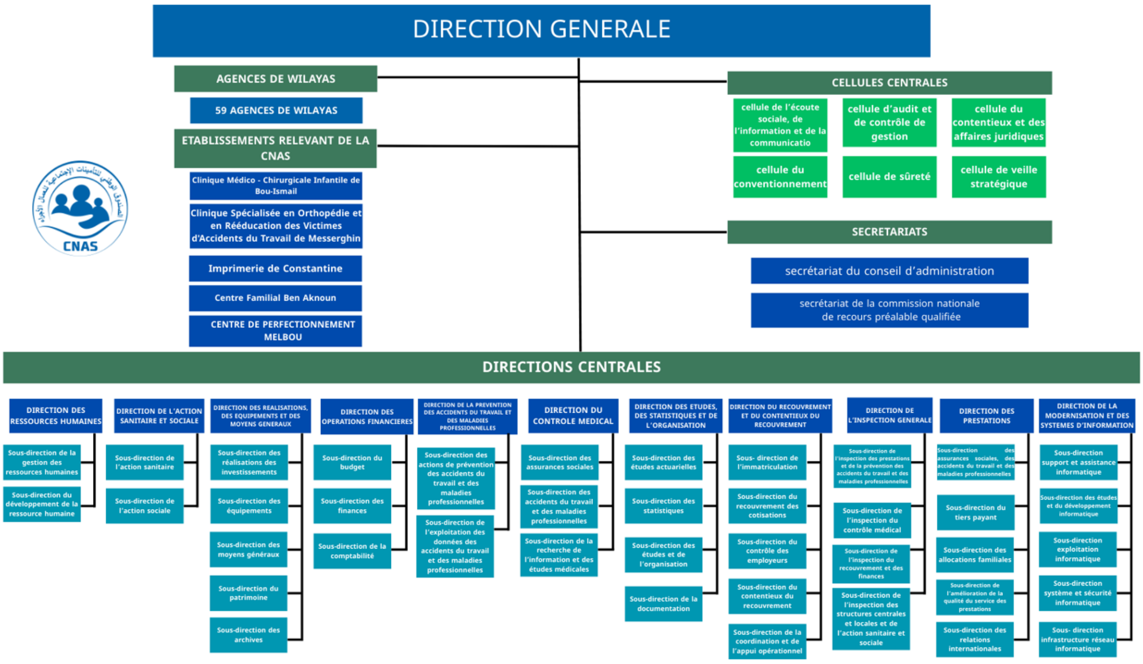

Figure 1.1: Structural organizational chart of CNAS

1.2.4 Beneficiaries

CNAS beneficiaries include salaried workers, apprentices, students, specific categories, and their dependents. This reminder highlights that the quality of benefits provided to the public is indirectly related to the quality of internal management and thus to the relevance of employee evaluation mechanisms.

1.2.5 Benefits

CNAS's main benefits cover healthcare, daily allowances, maternity insurance, disability pensions, death benefits, and occupational hazard coverage. This public service mission requires reliable, traceable, and consistent internal operation.

1.3 Missions and Roles of HRD and DMSI

1.3.1 Human Resources Department (HRD)

The **Human Resources Department (HRD)** is responsible for managing CNAS's human capital: personnel administration, recruitment, assignment, career monitoring, training, skills management, and, within the scope of this project, **performance and competency evaluation**.

Its main missions are:

- **Administrative management**: personnel file management, contracts, absences, leave, transfers, discipline;
- **Career management**: promotions, mobility, advancements, career path monitoring;
- **Training**: needs identification, training plan development, tracking and evaluation of actions;
- **Evaluation**: criteria definition, organization of evaluation campaigns, consolidation and exploitation of results;
- **HR Management**: production of dashboards, HR indicators, reports for General Management.

In this framework, HRD is not limited to administrative logic. It also participates in development objectives, clarification of professional expectations, and decision support, which aligns with research distinguishing several purposes of evaluation: administrative, developmental, role definition, and strategic [11].

1.3.2 Modernization and Information Systems Department (DMSI)

The **DMSI** plays a central role in implementing digitalization projects within CNAS. It is responsible for:

- the design, evolution, and maintenance of the information system;
- development, deployment, and support of business applications;
- management of hardware and software infrastructure;
- security of systems and data;
- supporting user departments in defining and implementing adapted solutions.

Within the scope of this thesis, the study and implementation of the solution takes place mainly within DMSI, in close functional coordination with HRD.

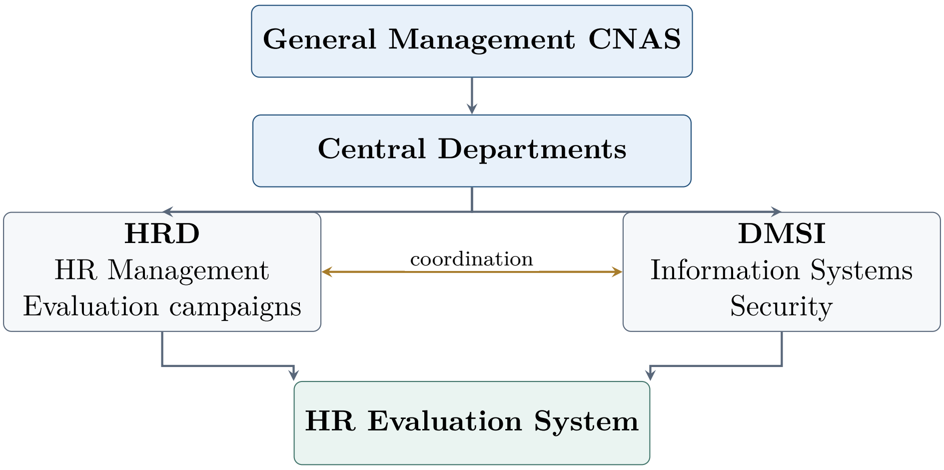

Figure 1.2: Positioning of HRD and DMSI in CNAS organization

1.4 Theoretical Framework and Concepts Related to the Field of Study

1.4.1 Information System (IS) and HRMS

An **information system** groups together all resources (software, data, procedures, hardware, and actors) allowing to collect, store, process, and distribute information. A **Human Resource Management System (HRMS)** is an information system specialized in HR management. It centralizes personnel information and automates processes such as administrative management, training, mobility, career monitoring, and evaluation.

In a modern logic of human resource management, HRMS does not only serve to archive data. It becomes a support for analysis, monitoring, coordination between actors, and decision support. In our case, it constitutes the technical foundation allowing to structure evaluation, secure its data, and improve its exploitation.

1.4.2 Competency

A **competency** designates a structured set of knowledge, skills, and behaviors that can be mobilized in a work situation. Within our project, we distinguish technical competencies, behavioral competencies, and, depending on positions, managerial competencies.

However, competency should not be viewed as an abstract or purely generic notion. Recent research on *competency models* shows that a credible system must be based on explicit, structured referentials adapted to the organization's context. In other words, competencies must be selected, organized, and linked to position families to make evaluation coherent, usable, and comparable within homogeneous areas [2]. This approach justifies, in our project, the establishment of differentiated referentials according to professions and evaluation profiles.

1.4.3 Performance

Performance reflects the level of objective achievement and the quality of work completed. Campbell and Wiernik show that it is multidimensional: it includes task performance, execution quality, discipline, cooperation, and, for certain positions, leadership dimensions or contextual contribution [3]. This vision moves us away from simplistic rating and justifies the use of structured grids, weighted criteria, and multi-indicator dashboards.

Within CNAS's context, this conception is particularly important. Professions are varied, constraints differ from one department to another, and performance cannot be reduced to simple quantitative output. It must be appreciated through results, but also through professional behaviors, procedural compliance, cooperation, and execution quality.

1.4.4 Competency and Performance Evaluation

Evaluation is a structured process aimed at:

- measuring individual performance over a given period;
- assessing competency levels and their evolution;
- identifying training and support needs;
- proposing development plans.

DeNisi and Murphy recall that evaluation should not be reduced to an annual rating, but thought of as a performance management system articulating objectives, feedback, measurement, development, and management [11, 1]. In this perspective, the logic of *performance management* goes beyond the act of evaluating to integrate a broader cycle comprising objective setting, regular monitoring, feedback production, results analysis, and employee development support.

This approach is particularly suited to an organization like CNAS, where evaluation cannot only serve to rank or punish, but must also support activity regulation, needs identification, and overall quality of HR management.

1.4.5 Feedback, Perceived Justice, and Biases

Research on feedback shows that its effects are not automatically positive. Some forms of feedback can improve performance, while others can degrade it if poorly designed. Moreover, recent work on continuous feedback shows that feedback mediated by a person, particularly by the hierarchical manager, produces more favorable effects on performance, motivation, and engagement than purely technological or impersonal feedback [7]. Literature also emphasizes that quantitative feedback and qualitative feedback do not have exactly the same effects: scores and indicators allow measurement and comparison, while comments and observations allow explanation, guidance, and support for improvement [7].

In an evaluation system, it is therefore relevant to combine:

- **quantitative feedback**, based on ratings, scores, and indicators;
- **qualitative feedback**, based on comments, observations, and assessments.

Furthermore, evaluation is exposed to several biases: leniency, severity, central tendency, halo effect, recency, or implicit comparison. In a multi-actor system, the perception of justice and process clarity thus play a central role in device acceptability. Research on *performance appraisal justice* indeed shows that a system perceived as just strengthens employee commitment, engagement, and indirectly their performance [12]. This idea justifies the integration, in our project, of traceability mechanisms, hierarchical validation, dispute resolution, and audit.

1.4.6 KPIs and Dashboards

KPIs (key performance indicators) allow managing HR activity: campaign completion rate, processing delays, volume processed, rework rate, competency scores, note dispersion, note inflation by manager, or gaps between observed and target competency. Provided documents show that an indicator is only relevant if it is defined, traceable, compared within a coherent area, and actionable.

In our project, KPIs are not viewed as mere display figures. They constitute management instruments allowing HRD, managers, and administration to identify gaps, track campaign progress, analyze results by position families, and support decision-making.

1.5 Compliance, Data Protection, and Profiling

HR evaluation processes personal data and may constitute profiling when used to analyze or predict work performance aspects. Law 18-07 and its 2025 amendment impose a compliance-by-design logic: designation of a DPO, maintenance of a treatment register, automated operations log, technical and organizational security measures, and strict access control [8, 9]. This has a direct translation into system architecture:

- role-based and scope-based access control;
- logging of consultations, modifications, and exports;
- data and transfer security;
- maintenance of usable traceability;
- minimum score explainability.

Beyond regulatory obligation, this compliance logic also strengthens system credibility, improves user trust, and supports procedural justice perception.

1.6 Project Scope

1.6.1 Implementation Framework

The project is developed in collaboration with:

- the **HRD**, which expresses business needs and validates referentials;
- the **DMSI**, which ensures technical study, design, and solution implementation.

1.6.2 Functional Scope

The proposed information system mainly covers:

- evaluation campaign management;
- definition and management of referentials;
- input, consultation, and validation of evaluations;
- centralization and historization of evaluation results;
- generation of indicators and reports;
- user, role, and access rights management.

1.6.3 Analytical Scope

Provided documents insist on the need to distinguish position families. Priority indicators are not the same for a file/counter-service position, an IT/DMSI position, an HR function, a managerial position, or a health/social structure. The system must therefore remain parameterizable, versioned, and capable of producing intra-family comparisons rather than cross-profession comparisons.

This adaptation principle aligns with research showing that an effective evaluation system must be sufficiently standardized to remain coherent, but sufficiently adaptable to account for organizational specificities, hierarchical levels, and different work contexts [1, 10].

1.7 Study of the Current Situation

1.7.1 General Findings

Before implementing a dedicated system, evaluations may rely on scattered supports (paper documents, Excel sheets, email exchanges) and non-standardized criteria. This situation creates several difficulties:

- **lack of uniformity** of criteria and evaluation grids;
- **absence of unique referential** of competencies;
- **limited traceability** of evaluations;
- **manual consolidation** of results, source of errors and wasted time;
- **difficulty in producing reliable statistics**.

These findings show that traditional evaluation practices, when based on scattered documents and manual processing, allow neither continuous performance monitoring, nor strategic data exploitation, nor satisfactory integration of feedback. They often produce results that are poorly comparable, difficult to audit, and weakly usable from HR decision perspective.

1.7.2 Current Process (Example of Workflow)

Generally, an evaluation process follows these steps:

- preparation of evaluation grids and criteria;
- communication to hierarchical managers and concerned services;

- conducting interviews and manual input of evaluations;
- transmission of results to HRD;
- consolidation and archiving;
- exploitation of results.

Information Flow

The following diagram illustrates a typical information flow before digitalization.

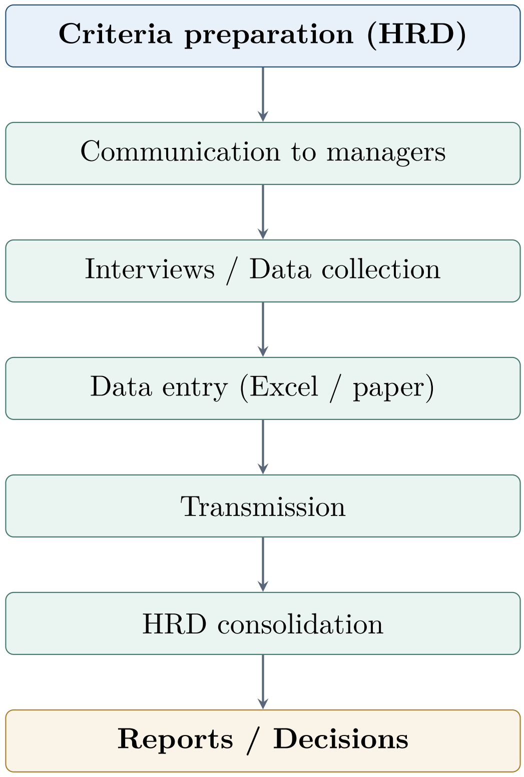

Figure 1.3: Information flow related to competency and performance evaluation process

1.8 Research Problem

Given the study of the current situation, the general research problem can be formulated as follows:

> *How to design and implement an information system allowing to standardize, automate, and secure the process of evaluating employee competencies and performance within CNAS, in order to improve analysis quality, decision traceability, and strategic management of human resources?*

The major difficulties to resolve are:

- the absence of a shared competency and criteria referential;
- the dispersion and centralization of evaluation data;
- the burden of manual consolidation;
- the difficulty in producing synthetic indicators;
- the need to integrate compliance and security constraints from the design phase.

This research problem is part of a broader dynamic of transforming evaluation practices towards integrated performance management systems, capable of responding to simultaneously operational, organizational, and strategic purposes.

1.9 Proposed Solution

To address this research problem, we propose the implementation of an **internal web application** dedicated to competency and performance evaluation within CNAS. The proposed solution notably includes:

- a centralized evaluation database;
- a structured referential of competencies and evaluation criteria;
- configurable grids according to position families;
- an input and validation workflow;
- dashboards and reports for HRD, managers, and administration;
- user and role management ensuring confidentiality;
- an audit log, treatment register, and complete operation traceability.

The proposed solution does not merely consist of digitizing an existing evaluation form. It aims to establish a structured logic of performance management, in which evaluation is connected to referentials, position profiles, indicators, a feedback process, and decision-making exploitation. In this sense, it responds to several purposes of evaluation: administrative purpose, development purpose, professional expectation clarification purpose, and strategic purpose [11].

1.9.1 Functional Diagram

The following figure illustrates, in simplified form, the functional diagram of the proposed system:

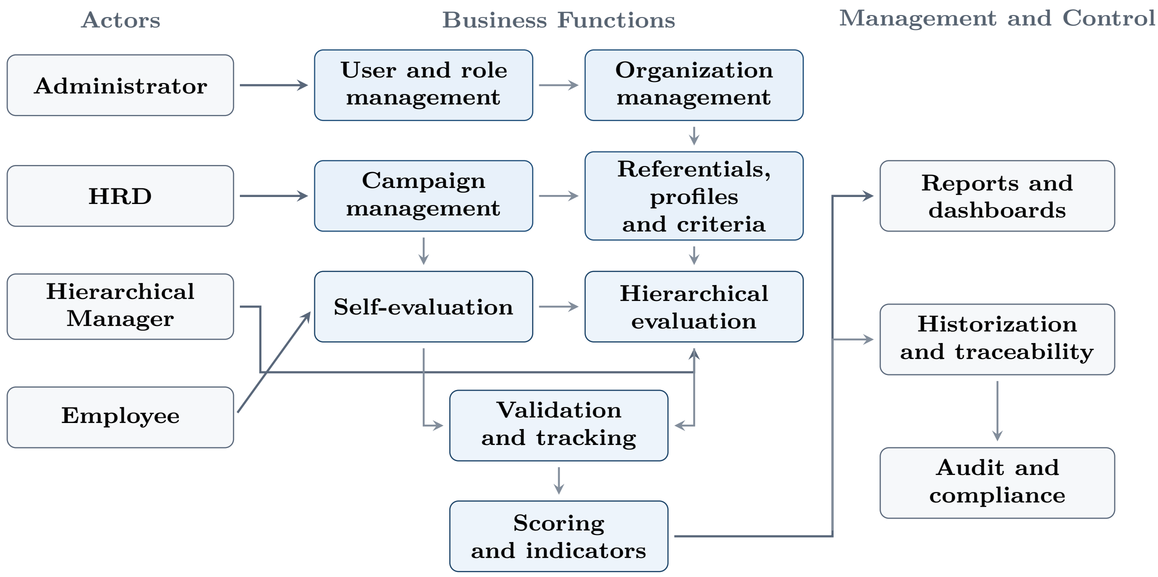

Figure 1.4: Overall functional diagram of the proposed evaluation system

1.10 Objectives

The main objectives of the project are:

- **Digitalize and automate** the evaluation process;
- **Standardize** evaluation criteria;
- **Centralize, secure and historize** HR data;
- **Improve traceability** and transparency;
- **Provide indicators** and reports facilitating decision support;
- **Reduce processing time** and limit errors;
- **Ensure compliance** of the system with data governance requirements.

Beyond these technical objectives, the project also aims to strengthen the consistency of the evaluation process, improve acceptability of the device by concerned actors, and create an exploitable basis for more structured management of competencies and performance within CNAS.

1.11 Conclusion

In this chapter, we presented CNAS and the strategic role of HRD and DMSI in human capital management and process digitalization. We studied the current situation regarding competency and performance evaluation, highlighted current limitations, and formulated the central research problem. Finally, we presented the project objectives and the general solution envisioned. The following chapter, *Analysis and Design*, builds on this foundation and translates it into functional, static, and dynamic modeling.

CHAPTER 2
ANALYSIS AND DESIGN

2.1 Introduction

In this chapter, we present the analysis and design of the proposed system. We begin with general specifications, which express the organizational, technical, and functional needs of the solution. Next, we detail the analysis and modeling choices using UML diagrams, to represent the system's static structure and dynamic behavior.

This chapter occupies a central place in the thesis, as it bridges the business need expressed in the previous chapter and the technical implementation presented in the following chapter. It also demonstrates that the proposed solution results not from an assembly of screens or tables, but from a coherent design approach oriented toward business, security, traceability, measurement quality, and exploitation of results.

2.2 General Description of the Solution

As a reminder, our solution consists of creating a web platform dedicated to evaluating employee competencies and performance within CNAS.

This platform is not limited to recording scores. It has been designed as a continuous performance management system, structured around several complementary dimensions: referential configuration, campaign management, evaluation input, self-evaluation, score calculation, validation tracking, indicator production, audit, compliance, and reporting. This orientation aligns with recent approaches where evaluation should be integrated into a broader *performance management* process, articulating objectives, measurement, feedback, tracking, and decision-making exploitation.

This platform mainly includes the following interfaces:

- Authentication interface;
- User and role management interface;
- Organization management interface (agencies, structures, services, employees);
- Evaluation campaign management interface;
- Profiles and evaluation criteria management interface;
- Evaluation input interface;
- Self-evaluation interface;
- Results consultation interface;
- Dashboards and reports interface;
- Audit and compliance interface.

These interfaces aim to:

- automate the evaluation process;
- centralize data related to employees, campaigns, and evaluations;
- define evaluation criteria and competency referentials;
- assign evaluations to hierarchical managers;
- allow input, validation, tracking, and historization of evaluations;
- integrate self-evaluation into the overall process;
- produce scores, indicators, reports, and dashboards;
- improve traceability, transparency, and exploitation of results;
- guarantee security, access control, and auditability of sensitive operations.

This solution is designed as a modular platform, where referentials, organization, campaigns, evaluations, scores, dashboards, and compliance functions are clearly separated. This separation facilitates maintenance, access control, audit, system scalability, and alignment with different evaluation purposes: administrative, developmental, strategic, and professional expectation clarification.

2.3 Requirements Specification

The purpose of this phase is to capture system context, identify technical and functional requirements, identify stakeholders involved in our system and their functions, and define the logical structure of the solution.

2.3.1 Organizational Needs

- Respect CNAS hierarchical organization.
- Improve task distribution among HRD, hierarchical managers, and employees.
- Ensure centralized and traceable management of evaluation campaigns.
- Facilitate tracking, consolidation, and exploitation of results.
- Provide differentiated functioning according to position families.
- Allow meaningful comparisons within homogeneous areas.
- Promote clarity of the evaluation process to improve acceptability.
- Integrate multiple evaluation purposes: HR management, decision support, development, and tracking.

These organizational needs are particularly important in CNAS's context, where profession diversity, hierarchical levels, and structure variety impose a system that is both consistent in structure and adaptable in application. Indeed, the same evaluation grid cannot be applied indiscriminately to a counter agent, an IT professional, an HR function, or a managerial position without producing weak or unfair comparisons.

2.3.2 Technical Requirements

These are determinative requirements for the system. Our system must guarantee:

- ease of use;
- secure authentication;
- access control according to roles and hierarchical areas;
- a centralized database ensuring information preservation and exploitation;
- an audit log tracing sensitive operations;

- a modular architecture facilitating system evolution;
- integration of a treatment register;
- form versioning logic;
- controlled and auditable exports;
- exploitable traceability of consultations, validations, corrections, and disputes.

The solution must also be capable of handling calculation mechanisms, historization, reporting, and data filtering according to roles. In this sense, it must be designed as an evaluation system, but also as a management and governance system for HR data.

2.3.3 Functional Requirements Capture

Identification of Actors

An actor represents any entity external to the application, whether a human user or another system, that directly interacts with it.

For our system, we identified the following actors:

- **Administrator**: manages users, roles, certain structure parameters, and supervises general platform administration;
- **HRD**: conducts campaigns, manages referentials, tracks results, exploits indicators, processes certain validations, and participates in compliance;
- **Hierarchical Manager**: evaluates employees in their area, consults self-evaluations, validates or rejects evaluations at hierarchical level, and tracks team activity;
- **Employee**: consults results, completes self-evaluation, tracks history, and can initiate certain requests, particularly in case of correction or dispute.

This actor distribution reflects multi-actor logic. It allows better responsibility distribution, improves information quality, and strengthens device justice perception. It aligns with inclusive logic, where multiple perspectives intervene in the evaluation cycle.

Functional Requirements

System functionalities are defined as the needs, actions, and reactions that the system must undertake in response to a primary actor's request.

The system must allow the following actors to:

- authenticate to the system;
- manage users;
- manage roles and access rights;
- manage agencies, structures, services, and employees;
- create an evaluation campaign;
- open, close, and track a campaign;
- define evaluation criteria;
- define evaluation profiles;
- define competency referential;
- assign evaluations to hierarchical managers;
- input evaluation scores and comments;
- save an evaluation as draft;
- submit an evaluation;
- validate or reject an evaluation according to responsibility level;
- complete a self-evaluation;
- consult evaluation history;
- consult individual and consolidated results;
- generate reports;
- export results;
- consult dashboards;
- record important actions in an audit log;
- manage notifications;
- manage disputes, corrections, or requests related to compliance when authorized;
- ensure traceability of validations, rejections, exports, and sensitive operations.

At business level, these functionalities must support not only evaluation recording, but also overall process tracking, results exploitation, feedback quality, and trust in the device.

2.3.4 System Architecture

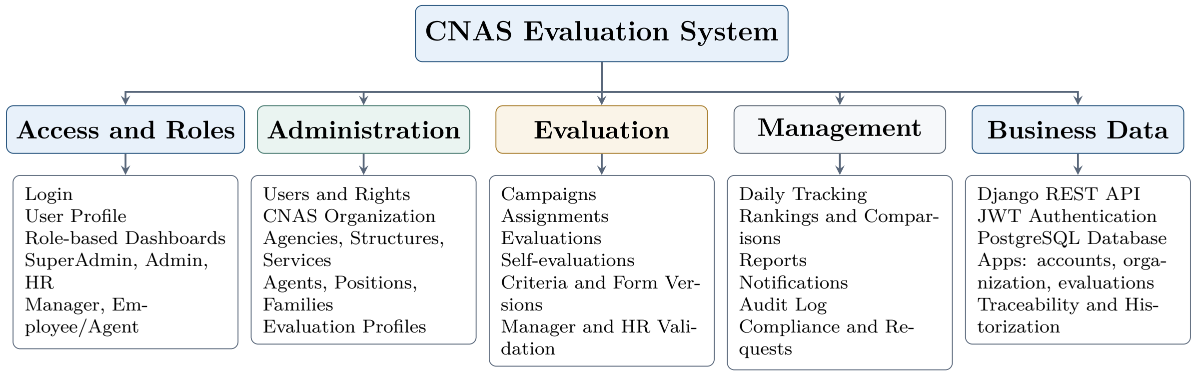

Figure 2.1: System Architecture

The system architecture reflects the modular logic actually retained in the application. It distinguishes access spaces by role, administration and organization modules, the core of the evaluation process, management functions, and data exposed by the Django REST API. This separation facilitates maintainability, readability, and responsibility distribution between React frontend, Django backend, and database.

2.3.5 Use Case Diagrams

A use case diagram represents a software system's functionalities from users' perspective. It shows interactions between actors and use cases, that is, the functionalities offered by the system [6, 17].

Use: define user needs, communicate with stakeholders, and design the system.

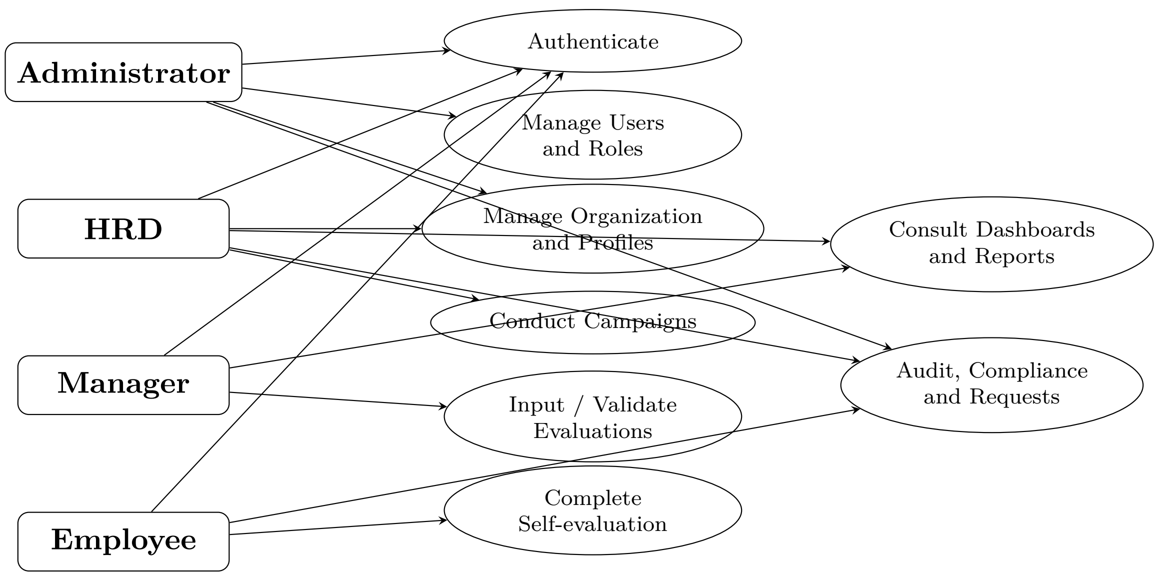

Figure 2.2: Use case diagram of the proposed evaluation system

The use case diagram shows that the system is not limited to simple "manager score result" relationship. It highlights a more complete chain: referential setup, campaign opening, evaluation input, self-evaluation, hierarchical validation, HRD exploitation, audit, and reporting. This logic is consistent with an integrated performance management approach.

Textual Documentation of Use Cases

The following table presents textual description of a central use case in the system: evaluation management. This case is central because it articulates several project dimensions: hierarchical attachment, evaluation referential, multi-criteria input, workflow, security, and traceability.

Table 2.1: Textual description of the "Manage an Evaluation" use case

Use Case	Manage an Evaluation
Actor	Hierarchical Manager
Objective	Evaluate an employee's performance and competencies according to criteria defined in an evaluation campaign and in the profile associated with their position.
Precondition	• User must be authenticated in the system; • An evaluation campaign must be active; • The employee must be assigned to the hierarchical manager; • An evaluation profile and criteria must be defined for the employee.
Main Scenario	1. The hierarchical manager accesses the list of employees to evaluate. 2. The system displays the corresponding evaluation form with applicable criteria and weightings. 3. The manager enters scores and comments according to evaluation criteria. 4. The system checks validity and consistency of entered information. 5. The manager saves the evaluation as draft or submits it. 6. The system saves data to the database. 7. The system calculates scores by criterion, by axis, and overall score according to weighting rules. 8. Sensitive actions are recorded in the audit log.
Postcondition	The evaluation is recorded in the system, with its status, scores, comments, and traceability.
Alternative Scenario	Some entered information is incorrect, inconsistent, or incomplete; the system displays an error message and requests correction. If rejected at a later level, the evaluation can be returned for revision.

2.4 Design

2.4.1 Functional Architecture

Provided documents and project analysis lead to retaining a modular architecture articulated around the following blocks:

- user and role management;
- hierarchical organization;
- competency referentials and criteria;
- evaluation profiles;
- evaluation campaigns;
- evaluations and self-evaluations;
- scoring and indicators;
- dashboards and reports;
- exports;
- compliance;
- audit log;
- notifications;
- versioning and historization.

This structuring is important for several reasons:

- it facilitates separation of functional responsibilities;
- it strengthens security and access control;
- it improves traceability;
- it prepares future system extension toward finer workflows, more advanced dashboards, and better data governance;
- it allows articulating administrative, development, management, and role clarification purposes.

Conceptually, this architecture expresses an important idea: evaluation is not an isolated module, but the meeting point between several subsystems, notably organization, referentials, roles, indicators, and compliance.

2.4.2 Class Diagram

A class diagram represents a system's static structure by showing classes, attributes, methods, and relationships between them. It allows visualizing key system concepts and their interactions [6, 17].

In our class diagram, main classes concern:

- users;
- roles;
- employees;
- organizational structures;
- evaluation campaigns;
- evaluation profiles;
- criteria;
- competencies;
- evaluations;
- responses or scores by criterion;
- audit logs;
- reports;
- notifications;
- compliance-related requests.

This diagram does not only represent a database. It translates essential business relationships: attachment of an employee to a service and manager, association of an evaluation profile to a position or job family, belonging of an evaluation to a campaign, link between scores, comments, history, and control.

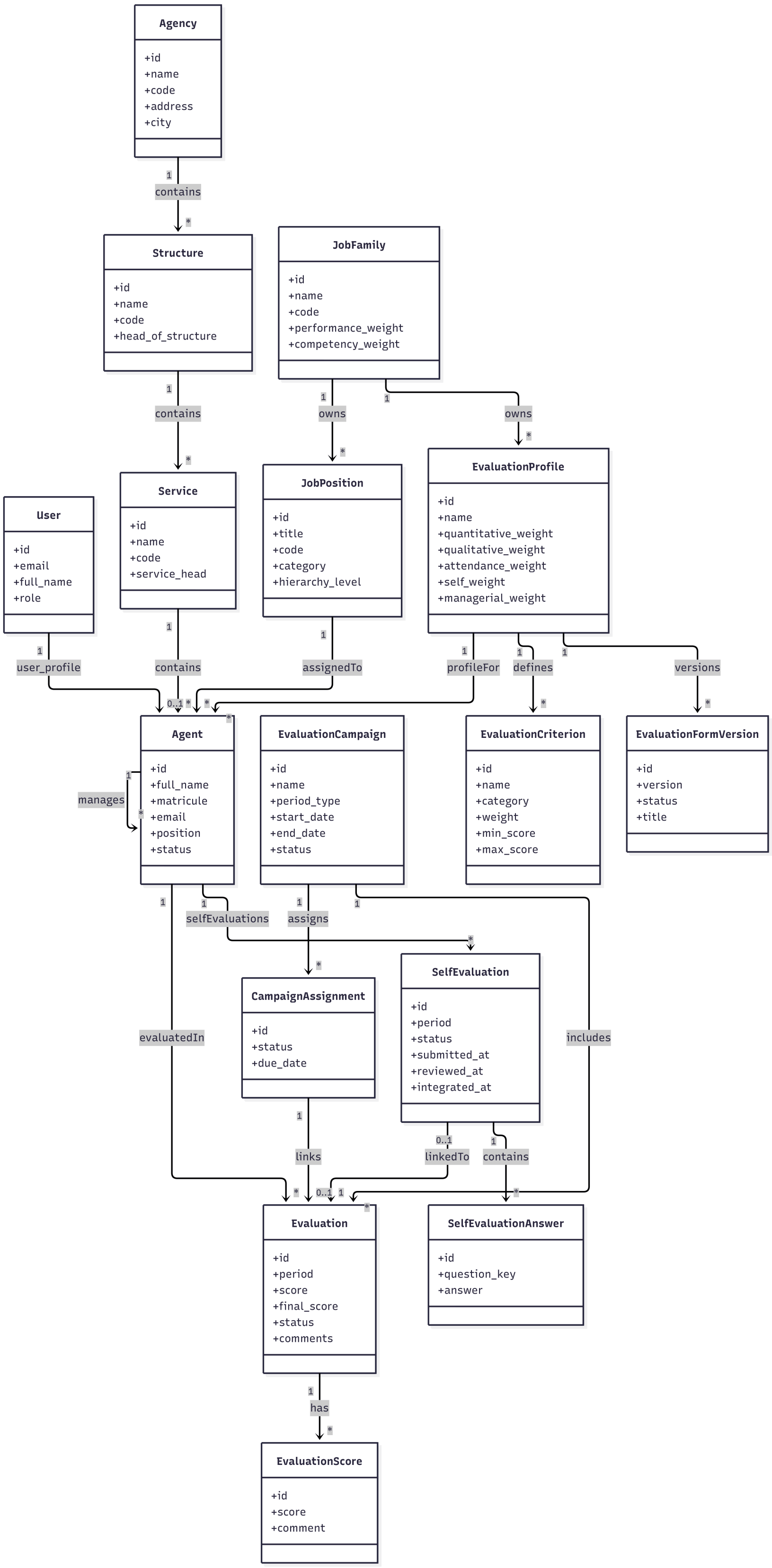

Figure 2.3: Class diagram of the employee skills and performance evaluation system

2.4.3 Sequence Diagrams

A UML sequence diagram is a visual tool representing interaction between objects in a system over time. It shows the chronological order of messages exchanged between objects to accomplish a particular task [6, 17].

The sequence diagrams retained in our work illustrate three central cases:

- campaign creation;
- evaluation input;
- evaluation validation.

Their interest is showing that the system is not only static. It functions according to sequence logic, controls, validations, and state changes, which is essential in a multi-actor evaluation process.

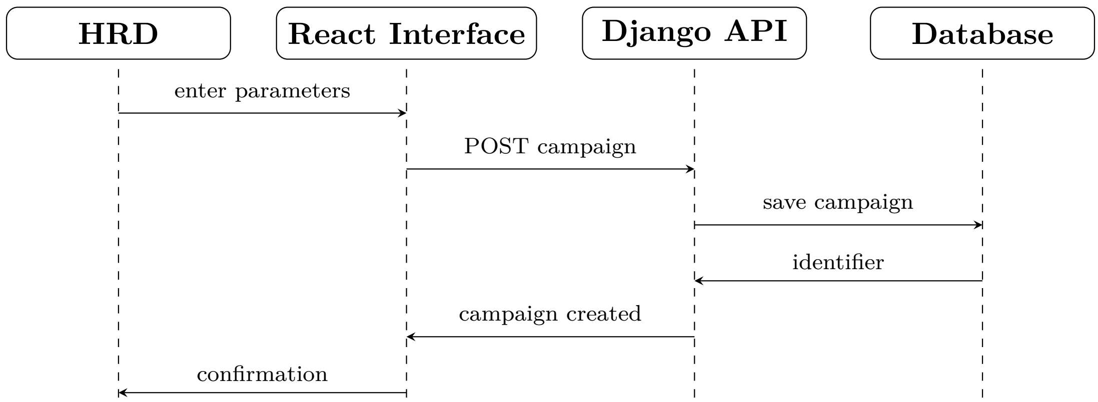

Figure 2.4: Sequence diagram for "Create an Evaluation Campaign" use case

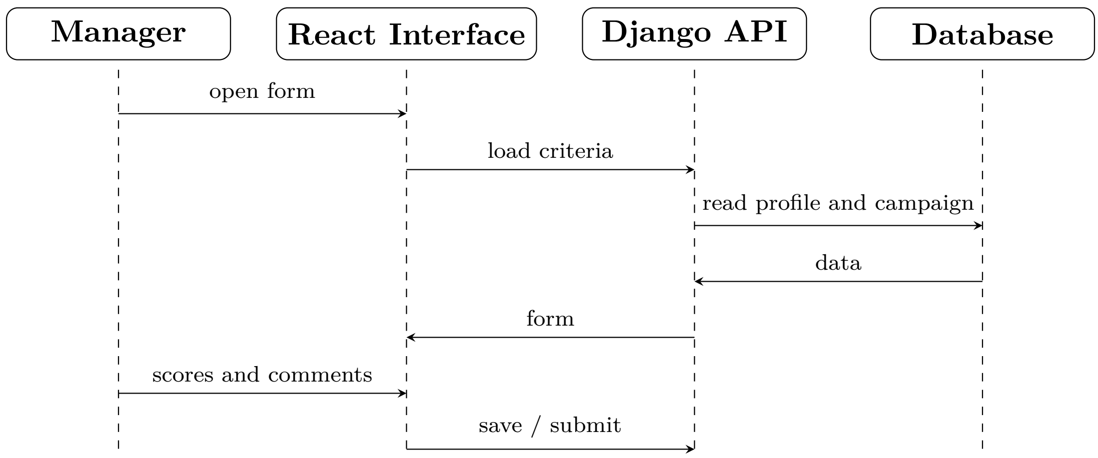

Figure 2.5: Sequence diagram for "Input an Evaluation" use case

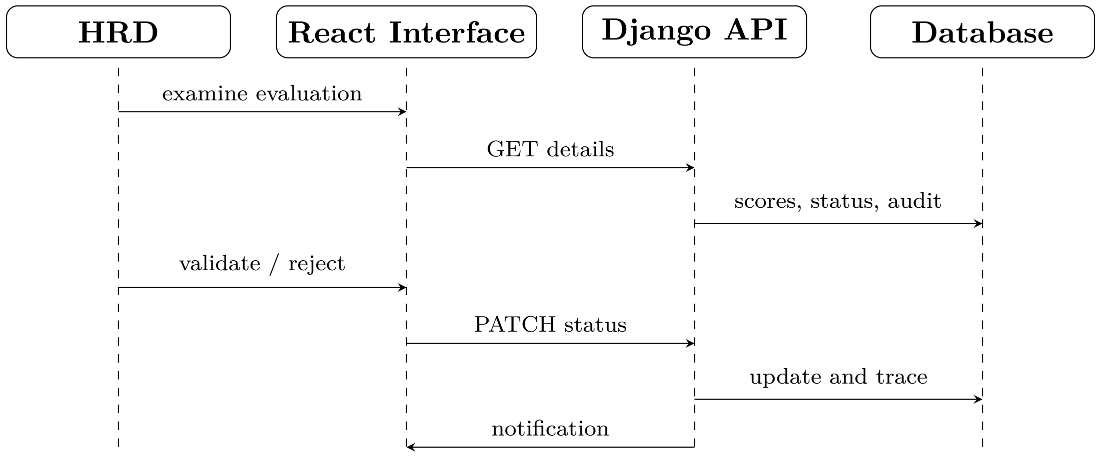

Figure 2.6: Sequence diagram for "Validate an Evaluation" use case

2.4.4 System Relational Model

To implement our database, we must convert the class diagram into a relational model.

The retained relational model must reflect not only main objects, but also the project's business logic: organizational hierarchy, referentials, profiles, campaigns, evaluations, self-evaluations, scoring, audit, and compliance.

Main System Tables

- **User** (<u>id_user</u>, name, email, password, id_role).
- **Role** (<u>id_role</u>, role_name).
- **Agency** (<u>id_agency</u>, name, code).
- **Structure** (<u>id_structure</u>, label, id_agency).
- **Service** (<u>id_service</u>, label, id_structure).
- **Employee** (<u>id_employee</u>, employee_id, name, firstname, position, service, manager, job_family, evaluation_profile).
- **JobFamily** (<u>id_family</u>, label, performance_weight, competency_weight).
- **EvaluationProfile** (<u>id_profile</u>, label, quantitative_weight, qualitative_weight, assiduity_weight, self_weight, managerial_weight).
- **Competency** (<u>id_competency</u>, label, description).
- **Criterion** (<u>id_criterion</u>, label, category, weight, min_note, max_note, id_profile).
- **EvaluationCampaign** (<u>id_campaign</u>, title, start_date, end_date, status).

- **Evaluation** (id_evaluation, evaluation_date, status, performance_score, competency_score, overall_score, comment, id_employee, id_campaign).
- **CriterionScore** (id_score, input_note, normalized_score, comment, id_evaluation, id_criterion).
- **SelfEvaluation** (id_self, status, average_score, id_employee, id_campaign).
- **Report** (id_report, type, generation_date, id_campaign).
- **Notification** (id_notification, content, creation_date, read, recipient).
- **AuditLog** (id_audit, action, timestamp, author, object, reason).
- **ComplianceRequest** (id_request, type, status, description, author, processing).

In the thesis, it is not necessary to list all technical tables from the actual project. However, it is essential to present the main tables that translate the system's functional logic.

2.4.5 Scoring System and Indicators

Provided resources show that a credible evaluation system must not stop at a raw score. It must at least distinguish performance score, competency score, and overall score, while explaining the main factors contributing to the result.

In our system, scoring logic relies on multiple levels:

- a **score by criterion**, obtained from the entered score;
- a **score by axis**, obtained by weighted aggregation of multiple criteria;
- a **performance score**, mainly based on quantitative, procedural, or assiduity dimensions;
- a **competency score**, based on qualitative, behavioral, self-evaluative, or managerial dimensions;
- an **overall score**, obtained by combining the two previous levels according to job family and evaluation profile.

This scoring architecture presents several advantages:

- it avoids reducing the process to a single score;
- it makes evaluation more explainable;

- it allows adapting weightings to position families;
- it distinguishes observed results from mobilized capacities and behaviors;
- it facilitates building analytical dashboards.

Indicators can include, depending on position families, volume processed, deadline compliance, cycle time, rework rate, file completeness, perceived feedback quality, procedural compliance, technical competency score, behavioral competency score, and leadership score.

However, these indicators must not be used without caution. They must be operationally defined, linked to a traceable data source, compared within a coherent area, and integrated into role-adapted dashboards. In particular, comparisons should be privileged within homogeneous populations rather than between very different professions.

2.4.6 Questionnaires and Measurement Quality

Support documents emphasize that a serious evaluation system must distinguish performance and competency, provide parameterizable forms, and rely on minimal questionnaire validation.

In practice, this means:

- review of items by subject matter experts;
- pretest on a small sample;
- acceptable internal consistency;
- documentation of the questionnaire used in each campaign;
- clear distinction between quantitative, qualitative, behavioral, and managerial criteria;
- explanation of weightings and calculation rules.

This logic improves device credibility and strengthens the thesis's scientific quality. It also allows avoiding two frequent errors: confusing competency and performance, and producing a final score without documenting its components.

Moreover, measurement quality is directly related to system acceptability. A well-constructed questionnaire, understandable, consistent with the evaluated position, and clearly linked to expected criteria reduces ambiguities, facilitates feedback, and strengthens the device's justice perception.

2.5 Conclusion

In this chapter, we presented the analysis and design of our competency and performance evaluation system within CNAS. We clarified organizational, technical, and functional needs, identified actors, formalized main use cases, then structured design around a modular architecture coherent with project requirements.

We also showed that the system is not based solely on technical logic, but on an articulated set of business choices: adaptation to position families, multi-actor workflow, explainable scoring, questionnaire quality, audit, compliance, and management by indicators. The mobilized scientific literature, as well as provided resources, allowed strengthening this approach by distinguishing performance, competency, feedback, perceived justice, and compliance. The next chapter presents concrete implementation of this design through application architecture, technologies used, and major system interfaces.

CHAPTER 3

IMPLEMENTATION

3.1 Introduction

In this chapter, after explaining in detail our solution's design, we address its implementation process. We focus on technical aspects associated with system creation, highlighting technology choices, modular organization of the application, security, and major interfaces.

3.2 Working Environment

In this section, we detail hardware and software tools as well as technologies used for development.

3.2.1 Technologies Used

React: React is a modern component-oriented JavaScript library used to build dynamic and responsive user interfaces. In our project, React was chosen to structure the interface in reusable components, improve frontend maintainability, and facilitate screen management dedicated to different system roles [13].

TypeScript: TypeScript is a typed superset of JavaScript that adds static typing, advanced autocompletion, and better safety during development. Its use in our application helps reduce errors, clarify data contracts exchanged with the API, and improve overall frontend code quality [14].

Python: Python is a versatile programming language, widely used for web development, automation, and data processing. Within the context of this application, Python forms the basis of backend development through its readability, productivity, and rich ecosystem [16].

3.2.2 Frameworks and Database

Django: Django is a high-level web framework based on Python. It provides a robust architecture for developing secure, maintainable, and rapidly deployable applications. In our project, Django was used to implement business logic, organize data models, manage authentication, and expose necessary services to the React interface [5].

PostgreSQL: PostgreSQL is an open source relational database management system, recognized for its robustness, SQL compliance, and reliability. It was chosen to ensure centralized storage of HR data, campaigns, evaluations, results, and traceability logs. Its relational model is particularly suited to the integrity, reporting, and historization needs of our system [15].

Chosen Technical Architecture: The realized solution relies on a modern client-server architecture: the user interface is developed with React and TypeScript, the back-end is implemented with Python and Django, and data is stored in PostgreSQL. This technology combination meets the modularity, maintainability, security, and scalability needs highlighted in the analysis reports provided for the thesis.

3.3 Application Architecture

The application is organized into several layers:

- presentation layer: web interfaces by role;
- business logic layer: authentication, campaigns, evaluations, scoring, reporting;
- data layer: users, employees, campaigns, criteria, responses, results, audits;
- compliance layer: audit log, exports, register, traceability.

This modular architecture allows separating navigation logic, business logic, and data persistence.

3.4 Web Application Security

Authentication: authentication ensures the truthfulness of a user's identity. In our system, its main objective is restricting access rights and confirming the user's right to access their work environment.

Beyond authentication, security relies on:

- role-based access control;
- filtering by hierarchical area;
- traceability of access and exports;
- maintenance of exploitable history;
- confidentiality precautions related to HR data.

3.5 Functional Organization by Role

The system must propose distinct views according to roles. This differentiation responds to a concrete business need and prevents a user from seeing data outside their scope.

3.5.1 Administrator Space

The administrator manages users, roles, assignments, and part of structure parameters.

3.5.2 HRD Space

HRD conducts campaigns, consolidates results, tracks global indicators, processes competency gaps, and prepares reports.

3.5.3 Hierarchical Manager Space

The manager evaluates employees in their area, tracks validations, and consults team dashboards.

3.5.4 Employee Space

The employee consults results, history, and, if the workflow allows, completes self-evaluation.

3.6 Presentation of Interfaces

3.6.1 Authentication

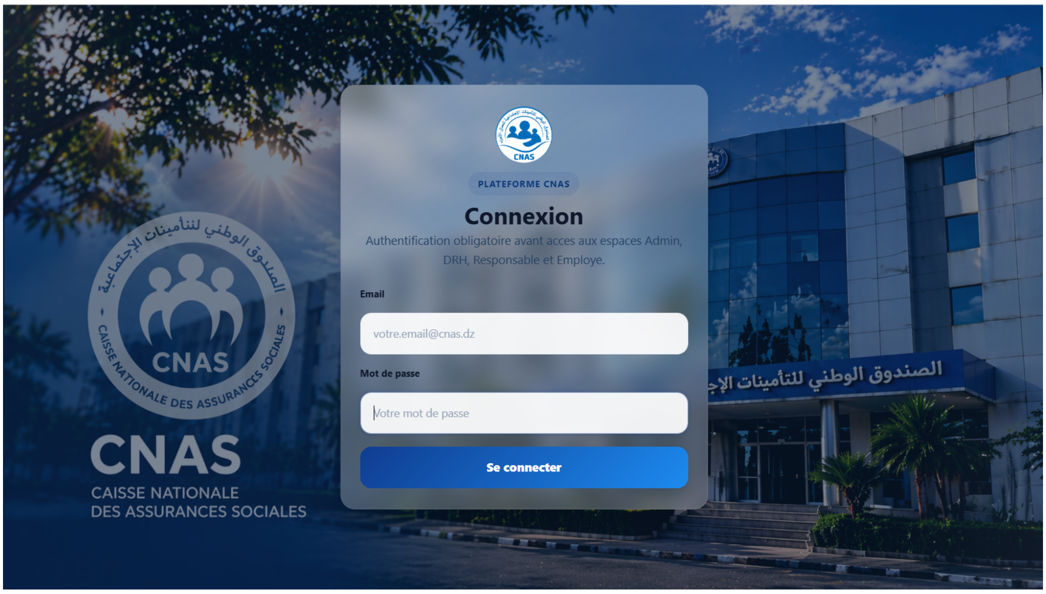

Figure 3.1: Authentication Interface

3.6.2 Home Page

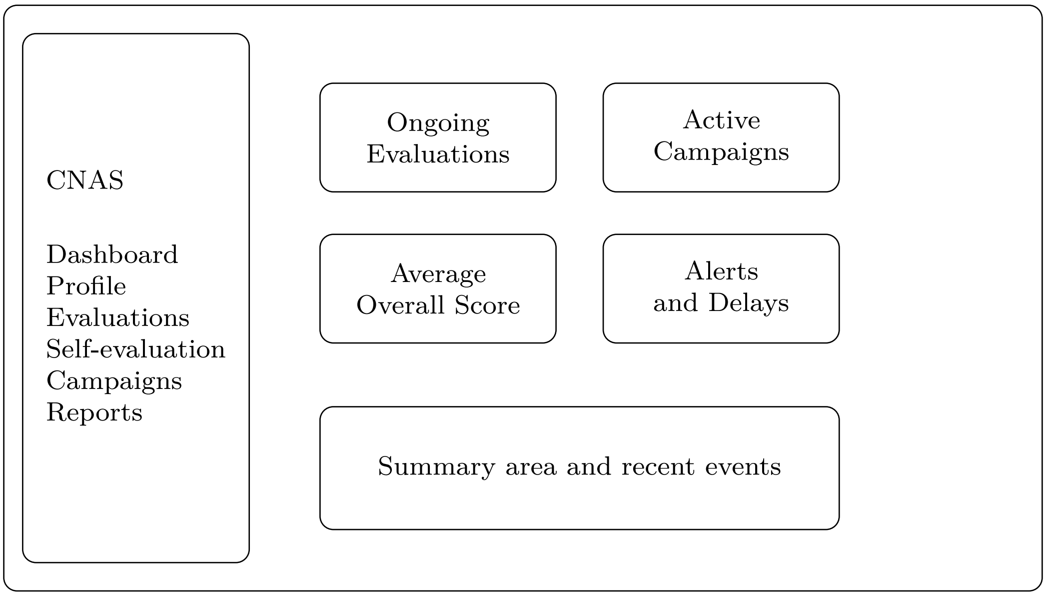

Figure 3.2: Home Page

3.6.3 Example Consultation Interface

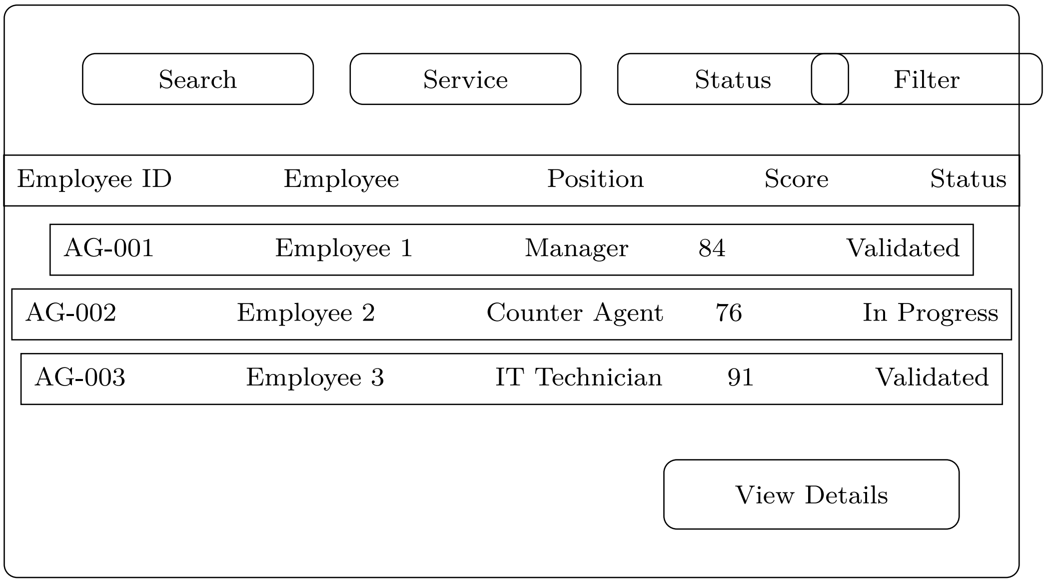

Figure 3.3: Example Consultation / Search Interface

3.7 Critical Discussion of Implementation

The achieved implementation demonstrates system feasibility. It already allows visualizing general role logic, interface organization, information centralization, and importance of access hierarchization. However, several enhancements remain possible: more advanced scoring engine, formal form versioning, dispute resolution, rich exports, more detailed dashboards, and better compliance industrialization.

3.8 Conclusion

In this chapter, we presented main technical choices, application organization, and representative interfaces. The achieved implementation constitutes a coherent foundation for a more complete and compliant HR evaluation system meeting functional and regulatory requirements.

GENERAL CONCLUSION

This thesis addressed the design and implementation of an information system dedicated to evaluating employee competencies and performance within CNAS. Through this work, we first analyzed the organizational context and identified limitations of existing practices: criteria heterogeneity, scattered supports, weak traceability, consolidation difficulty, and weak decision value of results.

We then showed that designing such a system cannot be reduced to simple numerical transposition of a paper form. It requires work framing position families, referentials, indicators, validation workflows, measurement quality, and data governance. The scientific literature mobilized, as well as provided resources, allowed strengthening this approach by distinguishing performance, competency, feedback, perceived justice, and compliance.

Finally, UML modeling, the relational model, modular architecture, and implementation choices allowed proposing a coherent, scalable system foundation adapted to CNAS needs. In perspective, this work could be extended by even more advanced dashboards, manager calibration mechanisms, better questionnaire formalization, dispute modules, and an even more complete compliance layer.

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APPENDICES

Appendix A: Comparative Table of Indicators

Indicator	Type	Operational Definition
Deadline Compliance	Q	Proportion of operations closed before set deadline.
Median Cycle Time	Q	Median duration between opening and closing of a file or processing.
Rework / Rework Rate	Q	Proportion of files returned or corrected.
File Completeness	Q	Proportion of complete files upon submission or validation.
Technical Competency Score	Qual→score	Weighted average of technical items related to position.
Behavioral Competency Score	Qual→score	Weighted average of cooperation, communication, and accuracy items.
Leadership Score	Qual→score	Aggregate of leadership and communication items for managers.
Perceived Feedback Quality	Qual→score	Perception of evaluation exchange by the evaluated person.
Note Inflation	Q	Abnormally high proportion of maximum scores assigned by an evaluator.
Note Dispersion	Q	Variance of scores assigned by an evaluator or service.

Appendix B: Position Families and Comparison Logic

Position Family	Priority Indicators	Recommended Comparison
Files / Counter / Payment Structures	volume, delays, cycle time, rework, completeness	intra-family and intra-service
IT / DMSI	ticket processing delay, incidents, availability, security compliance	intra-family
HR	campaign completion, HR processing delays, data quality, disputes	intra-family
Management	evaluation completion, team objectives, stability, note inflation/dispersion	intra-level
Health / Social Structures	quality incidents, protocol compliance, user satisfaction if available	intra-family

Appendix C: Figure References

PNG Images present in figures folder and used in thesis:

- ua1.png
- knas.png
- organigramme_cnas.png
- class_cnas.png
- authen1..png

Diagrams generated directly in LaTeX/TikZ:

- overall system functional diagram;
- system architecture;
- use case diagram;
- sequence diagrams;
- simplified mockups of home page and consultation/search.